lity & Margin Intelligence Model

Olist Brazilian E-Commerce | Commercial Analysis &

96,470	2,960	$15.4M	23.7%	8.1%	617
Delivered Orders	Active Sellers	Total Revenue	Avg Contribution Margin	Delivery Delay Rate	Renegotiation Targets

01 | BUSINESS PROBLEM

Olist, a Brazilian e-commerce marketplace, operates with thousands of third-party sellers across 27 states. As a Business Analyst, the challenge was to translate raw transactional data into clear commercial intelligence: **is revenue growth actually creating sustainable profit, or is the platform growing volume while quietly eroding margin?**

Stakeholders needed answers to five specific questions that directly inform commercial strategy: seller renegotiation priorities, logistics improvement targets, revenue risk assessment, and a clear action plan. This project structured the analysis around those questions — from data to decision.

02 | DATASET & APPROACH

Dataset	Olist Brazilian E-Commerce (Kaggle)
Records	96,470 delivered orders (filtered from 99,441 total)
Tables Used	orders, order_items, payments, customers, sellers, reviews, products
Period	2016–2018 (24 months of transaction data)
Scope	2,960 active sellers across 27 Brazilian states

8-Phase Methodology:

Phase 1: Data Foundation — joined 7 tables into a clean master dataset (96,470 rows, zero duplicates)

Phase 2: Financial Modeling — defined revenue, simulated COGS (70% of price), calculated profit and contribution margin per order

Phase 3: Seller Profitability Intelligence — ranked and segmented all 2,960 sellers into a 2x2 matrix

Phase 4: Operational Impact Analysis — calculated delivery delays by seller and region; correlated with financial performance

Phase 5: Revenue Stability & Risk — measured monthly volatility (CV), rolling trends, and concentration ratios

Phase 6: Margin Risk Scoring — built a composite risk score (weighted: 40% margin + 35% delay + 25% concentration)

Phase 7: Dashboard & Reporting — designed multi-sheet Excel workbook with charts, tables, and KPI strips

Phase 8: Strategic Recommendations — translated all findings into a 5-question executive action plan

03 | KEY FINDINGS

FINDING 1 — Margin Erosion Across 617 Sellers

617 sellers (21% of the base) are classified as HIGH REVENUE + LOW MARGIN — generating $5.2M in revenue at only 22.2% average contribution margin. Star-segment sellers (same revenue tier) achieve 26.4% margin. Bringing Risk sellers to Star-level margins would generate an additional **$151,677 in profit** — a 12% increase in segment profitability.

- Risk sellers operate 4.2 percentage points below the Star seller margin benchmark
- 617 sellers flagged for renegotiation, with clear revenue-ranked priority list
- Root causes identified: low-price product mix, excessive freight relative to order value

FINDING 2 — Delivery Delays Destroy Customer Satisfaction

8.1% of all orders experienced delivery delays. The financial and reputational impact is significant: delayed orders receive an average review score of **2.57 vs. 4.29** for on-time deliveries — a 40% drop in customer satisfaction.

- 7,826 orders delayed, representing $1.35M in revenue carried with high churn risk
- Estimated revenue at risk from delay-driven churn: $67,590 (5% conservative estimate)
- Highest delay rates concentrated in specific seller states — regional action required

FINDING 3 — Revenue Stability is Moderate, Not Fragile

Monthly revenue Coefficient of Variation is 0.59 — classified as Moderate. The platform shows clear seasonal peaks (Nov 2017: $1.15M peak) and is not overly concentrated: Top 10 sellers represent only 12.9% of revenue, limiting single-seller dependency risk.

- Revenue range: $300K–$1.15M monthly across the analysis period
- Top 20 sellers control 21.1% of revenue — manageable concentration
- Proactive diversification recommended before stability tips toward volatile

04 | VISUALIZATIONS

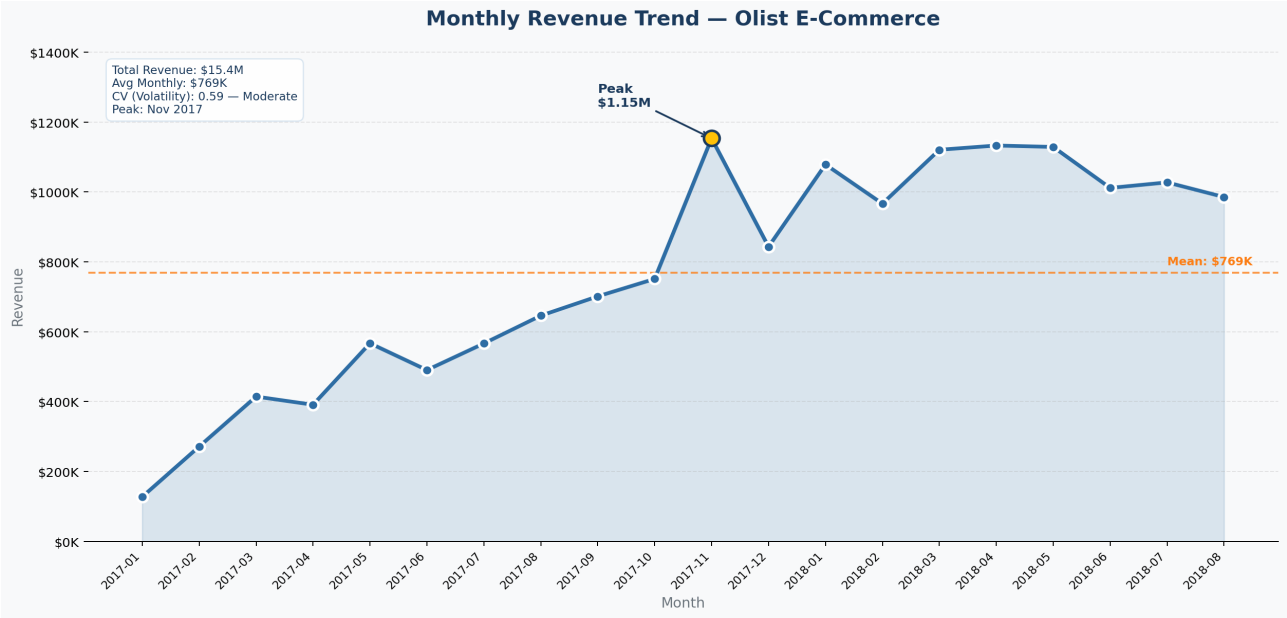

Fig 1 — Monthly Revenue Trend: 24-month trajectory with mean baseline and peak annotation

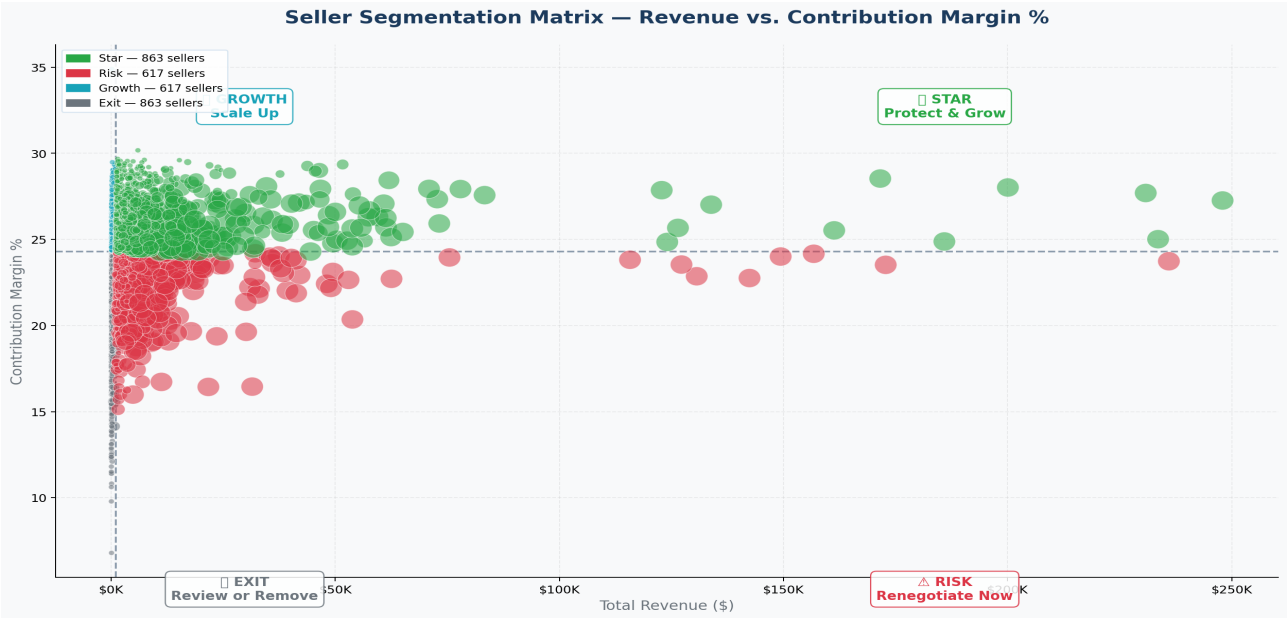

Fig 2 — Seller Segmentation Matrix: 2,960 sellers plotted by Revenue vs Contribution Margin %

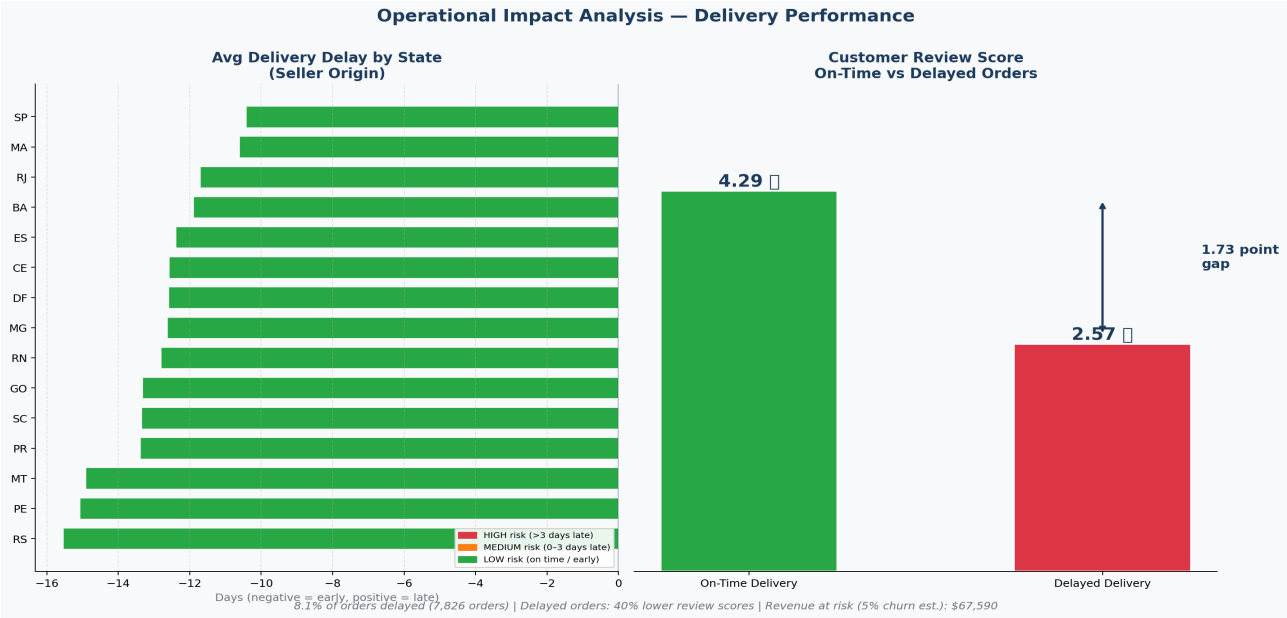

Fig 3 — Operational Impact: Regional delay by state + customer review score comparison

05 | STRATEGIC RECOMMENDATIONS

IMMEDIATE (0–30 days)	Launch renegotiation with the top 50 Risk-segment sellers ranked by revenue. Enforce a minimum 25% contribution margin floor as a platform listing requirement.
SHORT-TERM (30–90 days)	Deploy seller-level delivery SLA monitoring. Sellers exceeding 15% delay rate face listing restrictions. Prioritize logistics investment in the 5 highest-delay states.
MEDIUM-TERM (3–6 months)	Rebuild seller mix toward Star and Growth segments. Launch targeted seller acquisition program in high-margin product categories.
LONG-TERM (6–12 months)	Target 30%+ blended contribution margin and sub-5% delay rate. Establish a monthly Revenue Quality Council using this model as the standing dashboard.

06 | TOOLS & SKILLS DEMONSTRATED

Business Analysis	Requirements definition, stakeholder question framing, insight-to-action translation
Financial Modeling	Contribution margin simulation, profitability analysis, cost structure decomposition
SQL	Multi-table joins, window functions (RANK, PERCENTILE_CONT), CTEs, analytical views
Excel (Advanced)	Multi-sheet dashboards, KPI design, conditional formatting, formula-driven reporting
Data Visualization	Matplotlib / seaborn — trend charts, segmentation matrix, operational impact charts
Strategic Reporting	Seller segmentation, composite risk scoring, executive recommendation documents